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PROACTIVE DSDS HANDOFF UNDER GRANT SUSPENSION FOR WIRELESS NETWORKS

Abstract

When a first (e.g., CBRS) network receives instructions to modify its operations with a dual-SIM, dual-subscription (DSDS) device, e.g., due to the local incumbent activity, instead of modifying CBRS operations for the DSDS device, the DSDS device is proactively handed off to a second (e.g., non-CBRS) network, thereby avoiding a reduction of quality of service to the DSDS device. If and when the local incumbent activity ceases, the original CBRS operations can be automatically restored by handing off from the non-CBRS network back to the CBRS network. In some implementations, these functions are orchestrated by a DSDS Mobile Transfer Entity of the CBRS network.

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Background/Summary

BACKGROUND

Field of the Disclosure

[0001] The present disclosure relates to communication networks and, more specifically but not exclusively, to CBRS communication networks.

Description of the Related Art

[0002] This section introduces aspects that may help facilitate a better understanding of the disclosure. Accordingly, the statements of this section are to be read in this light and are not to be understood as admissions about what is prior art or what is not prior art.

[0003] A dual-SIM, dual-subscription (DSDS) wireless device is able to communicate with two different wireless networks at the same time. A DSDS device has a connection manager that controls the handing off of connections between the two wireless networks.

[0004] In a Citizens Broadband Radio Service (CBRS) network, incumbent (i.e., government) users have priority of non-incumbent users, such as commercial wireless network providers. When an active incumbent user is present, the Spectrum Access System (SAS) for the CBRS network will instruct non-incumbents to modify their use of the CBRS spectrum to avoid interfering with the incumbent. Modification may involve a network provider's base stations (i.e., CBSDs) temporarily (i) moving existing communications with its customers' wireless user equipments (UEs) from one CBRS sub-band to another CBRS sub-band, (ii) significantly reducing transmit power levels, and/or (iii) suspending operations altogether. If and when the incumbent's local use of the CBRS spectrum terminates, the SAS may instruct the non-incumbents to resume their previous use of the CBRS spectrum. These temporary modifications may result in decreased capability of the non-incumbents to support their customers in a satisfactory manner.

SUMMARY

[0005] Problems in the prior art are addressed in accordance with the principles of the present disclosure by pre-emptively and automatically handing off DSDS communications from a CBRS network to a different wireless network when the CBRS network's SAS instructs the CBRS network to modify its communications due to the presence of a locally active incumbent. If and when the incumbent is no longer locally active, the original DSDS communications with the CBRS network can be automatically resumed.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] Embodiments of the disclosure will become more fully apparent from the following detailed description, the appended claims, and the accompanying drawings in which like reference numerals identify similar or identical elements.

[0007] FIG. 1 is a simplified diagram representing an example configuration in which two different networks—a CBRS network and a non-CBRS network—operate in the same geographic region and have at least some overlapping coverage areas;

[0008] FIG. 2 is a block diagram of some of the infrastructure supporting the configuration of FIG. 1;

[0009] FIG. 3 is a message flow diagram of an example sequence of messages for the handoff of communications of the DSDS device of FIG. 2 from the CBRS network to the non-CBRS network;

[0010] FIG. 4 is a message flow diagram of an example sequence of messages that may occur for the handoff of communications of the DSDS device of FIG. 2 from the non-CBRS network back to the CBRS network after the incumbent's local activity has ceased; and

[0011] FIG. 5 is a simplified hardware block diagram of an example node that can be used to implement any of the nodes of FIGS. 1-4.

DETAILED DESCRIPTION

[0012] Detailed illustrative embodiments of the present disclosure are disclosed herein. However, specific structural and functional details disclosed herein are merely representative for purposes of describing example embodiments of the present disclosure. The present disclosure may be embodied in many alternate forms and should not be construed as limited to only the embodiments set forth herein. Further, the terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting of example embodiments of the disclosure.

[0013] As used herein, the singular forms “a,” “an,” and “the,” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It further will be understood that the terms “comprises,” “comprising,” “contains,” “containing,” “includes,” and/or “including,” specify the presence of stated features, steps, or components, but do not preclude the presence or addition of one or more other features, steps, or components. It also should be noted that in some alternative implementations, the functions/acts noted may occur out of the order noted in the figures. For example, two figures shown in succession may in fact be executed substantially concurrently or may sometimes be executed in the reverse order, depending upon the functions/acts involved.

[0014] FIG. 1 is a simplified diagram representing an example configuration **100** in which two different networks—a CBRS network and a non-CBRS network—operate in the same geographic region and have at least some overlapping coverage areas. As shown in FIG. 1, towers **110** represent the CBSDs of the CBRS network, where each CBSD has a corresponding coverage area **112**, while tower **120** represents the base station of the non-CBRS network having a corresponding, relatively large coverage area **122** that covers a number of the CBSD coverage areas **112**.

[0015] FIG. 2 is a block diagram of some of the infrastructure **200** supporting the configuration **100** of FIG. 1. In particular, FIG. 2 shows the core networks **210** and **220** for the CBRS and non-CBRS networks, respectively, of FIG. 1. In addition, FIG. 2 shows (i) the Operation and Support System (OSS) **212** of the CBRS network which communicates with the CBSD network's SAS **230** and (ii) a DSDS Mobile Transfer Entity (DMTE) **214**, which supports the pre-emptive handoff function of the present disclosure. Depending on the particular implementation, the DMTE **214** may be implemented as part of or separate from the OSS **212**, including possibly in the cloud. FIG. 2 also shows a DSDS device **240** communicating with the CBRS core network **210** via one of the CBRS network's CBSDs (not explicitly shown in FIG. 2).

[0016] According to conventional CBRS network operations, when the SAS **230** detects the presence of an incumbent (not shown in FIG. 2), the SAS **230** instructs the OSS **212** to modify the operations of the CBRS network. The SAS **230** typically gives the CBRS network a specified amount of time (e.g., 60 seconds) to modify its operations to avoid interfering with the incumbent.

[0017] According to certain embodiments of the present disclosure, the OSS **212** shares the SAS instructions with the DMTE **214**, which orchestrates a handoff of the communications between the DSDS device **240** and the CBRS network to the non-CBRS network, such that the communications between the DSDS device **240** and the non-CBRS network will instead be handled by the non-CBRS core network **220**. If and when the incumbent's local activity ceases and the SAS **230** instructs the OSS **212** to resume the CBRS network's previous use of the CBRS spectrum, the DMTE **214** can orchestrate a handoff of the communications between the DSDS device **240** and the non-CBRS network back to the CBRS network, such that the communications between the DSDS device **240** and the CBRS network will once again be handled by the CBRS core network **210**, as shown in FIG. 2.

[0018] FIG. 3 is a message flow diagram of an example sequence **300** of messages for the handoff of communications of the DSDS device **240** of FIG. 2 from the CBRS network to the non-CBRS network. As shown in FIG. 3, the sequence **300** begins at step **306** with the DSDS device **240** having an on-going data session with a CBSD **302** of the CBRS network.

[0019] In step **308**, after detecting local incumbent activity, the SAS **230** transmits a grant suspension message to the CBRS network's OSS **212** of FIG. 2 identifying which of the CBRS

network's CBSDs (including, in this example, the CBSD **302**) need to modify their operations within a specified period of time (e.g., 60 seconds). In response, in step **310**, the OSS **212** forwards the grant suspension message to the CBSD **302** (and any other affected CBSDs) instructing the CBSD **302** to modify its operations as dictated by the SAS **230**, but to wait until after a specified period of time (e.g., slightly less than 60 seconds) has passed (to give time for any proactive handoffs to occur).

[0020] In addition, in step **312**, the OSS **212** shares the grant suspension message with the DMTE **214**. In response, in step **314**, the DMTE **214** analyzes the coverage areas (i.e., cells) that are affected by the grant suspension message (including, in this example, the cell associated with the CBSD **302**) to determine information needed to hand off certain communications (if any) to the non-CBSD network. This information includes an identification of the cells impacted by the grant suspension and whether the grant suspension affects all of the cells or only some of them.

[0021] In step **316**, the DMTE **214** transmits a request to the CBRS core **210** for the identities of any DSDS devices currently located within the cells affected by the grant suspension, and, in step **318**, the CBRS core **210** responds with the identities of those DSDS device (including, in this example, the DSDS device **240**). In some implementations, for each DSDS device, the information transmitted in step **318** includes one or more of: [0022] The International Mobile Subscriber Identity (IMSI) stored in the device's Subscriber Identify Module (SIM) associated with the CBRS network; [0023] The Global Unique Temporary Identifier (GUTI), a core network temporary identifier used for access security context; and [0024] The Cell Radio Network Temporary Identifier (C-RNTI) that identifies the device's Radio Resource Control (RRC).

[0025] In step **320**, the DMTE **214** analyzes the data received from the CBRS core **210** in step **318** to determine which of those DSDS devices identified in step **318** have currently active communication sessions (including, in this example, the DSDS device **240**) and which of those identified DSDS devices are currently inactive.

[0026] In step **322**, the DMTE **214** transmits a request to the CBRS core **210** for information about those DSDS devices having active communication sessions (including, in this example, the DSDS device **240**), and, in step **324**, the CBRS core **210** responds with the requested information. In some implementations, for each active session of a DSDS device, the information transmitted in step **324** includes one or more of: [0027] The Packet Data Unit (PCU) session identifier; [0028] The Evolved Packet System (EPS) bearer; [0029] The Quality of Service (QOS) Flow Identifier (QFI); and [0030] The Tunnel End Point Identifier (TEID).

[0031] In step **326**, the DMTE **214** instructs the CBRS core **210** to initiate handoff procedures for specific DSDS devices at specific CBSDs (including, in this example, the DSDS device **240** at CBSD **302**). In response, in step **328**, the CBRS core **210** transmits a message to the DSDS device **240** via the CBSD **302** instructing the DSDS device **240** to initiate a communication session between the DSDS device **240** and the non-CBRS network.

[0032] In step **330**, the DSDS device **240** (actually the connection manager (not shown) of the DSDS device **240**) prepares the DSDS device **240** to initiate a duplicate session with the non-CBRS network. As understood by those skilled in the art, in this context, a duplicate session is a redundant communication session (i.e., having the same data) between the DSDS device **240** and the non-CBRS network that occurs in parallel with the already existing communication session between the DSDS device **240** and the CBRS network of step **306**.

[0033] To that end, in step **332**, the DSDS device **240** communicates with a base station of the non-CBRS network (in this example, gNodeB **304**) to initiate a new data session. In step **334**, the DSDS device **240** merges the new non-CBRS data session with the existing CBRS data session, such that, in step **336**, the duplicate session with the non-CBRS network is established.

[0034] In step **338**, having established the duplicate session with the non-CBRS network's gNodeB **304**, the DSDS device **240** stops transmitting data to the CBRS network's CBSD **302** such that, in step **340**, the communication session with the CBRS network is terminated. In some possible

implementations, the connection manager of the DSDS device **240** automatically terminates the session with the CBRS network after recognizing that existence of the two duplicate sessions. In other possible implementations, the DSDS device **240** transmits a message to the CBRS network confirming the establishment of the duplicate session with the non-CBRS network and the CBRS network responds by instructing the DSDS device **240** to terminate the session with the CBRS network.

[0035] Note that the establishment of the duplicate session with the non-CBRS network and the termination of the original session with the CBRS network in FIG. **3** all occur within the duration (e.g., 60 seconds) specified by the SAS **230** in step **308** for modifying the communications of the CBRS network. In this way, the DMTE **214** pre-empts the conventional modification of CBRS network operations for impacted DSDS devices by proactively handing off those communications to the non-CBRS network, thereby avoiding at least some of the conventional decrease in capability of the CBRS network due to incumbent activity and provide at least those DSDS customers with high-quality service.

[0036] As described previously, if not terminated for other, conventional reasons, the new communication session between the DSDS device **240** and the non-CBRS network may continue as long as the incumbent's local activity continues.

[0037] FIG. **4** is a message flow diagram of an example sequence **400** of messages that may occur for the handoff of communications of the DSDS device **240** of FIG. **2** from the non-CBRS network back to the CBRS network after the incumbent's local activity has ceased. As shown in FIG. **4**, the sequence **400** begins at step **402** with the DSDS device **240** having the same on-going data session with the gNodeB **304** of the non-CBRS network that was established by the sequence **300** of FIG. **3**.

[0038] In step **404**, after determining that the incumbent's local activity has ceased, the SAS **230** transmits a grant restoration message (and associated heartbeats) to the CBRS network's OSS **212** identifying which of the CBRS network's CBSDs (including, in this example, the CBSD **302**) need to modify their operations within a specified period of time (e.g., 60 seconds). In response, in step **406**, the OSS **212** forwards the grant restoration message to the CBSD **302** instructing the CBSD **302** to modify its operations after a specified period of time (e.g., slightly less than 60 seconds).

[0039] In addition, in step **408**, the OSS **212** shares the grant restoration message with the DMTE **214**. In response, in step **410**, the DMTE **214** analyzes the cells that are affected by the grant restoration message (including, in this example, the cell associated with the CBSD **302**) to determine information needed to hand off communications (if any) from the non-CBSD network back to the CBSD network. This information includes an identification of the cells having a restored heartbeat, an identification of the DSDS devices that were present in the CBSD network before the grant suspension was imposed, and a determination of the projected load associated with the restoration of service to those DSDS devices.

[0040] In step **412**, the DMTE **214** transmits a request to the CBRS core **210** for the identities of any DSDS devices currently located within the cells affected by the grant restoration, and, in step **414**, the CBRS core **210** forwards the request of step **412** to the CBSD **302**. In step **416**, the CBSD **302** transmits a page message to determine which UEs are currently located within its cell. In step **418**, the DSDS device **240** transmits a page response to the CBSD **302**, which, in step **420**, the CBSD **302** forwards to the CBRS core **210**.

[0041] In step **422**, in response to the page response of step **420** and the request of step **412**, the CBRS core **210** provides information about the UEs (including, in this example, the DSDS device **240**) to the DMTE **214**. In some implementations, step **422** involves the same types of information as for step **318** of FIG. **3**.

[0042] In step **424**, the DMTE **214** analyzes the data received from the CBRS core **210** in step **422** to identify those identified DSDS devices that have currently active communication sessions with the CBRS network and those identified DSDS devices that do not (including, in this example, the

DSDS device **240**).

[0043] In step **426**, the DMTE **214** instructs the CBRS core **210** to establish a connection with specific DSDS devices at specific CBSDs (including, in this example, the DSDS device **240** at CBSD **302**) and retrieve information about those DSDS devices having active communication sessions (including, in this example, the DSDS device **240**). In step **428**, the CBRS core **210** forwards those instructions to the CBSD **302**, which, in step **430**, establishes the connection with the DSDS device **240** and requests the session information.

[0044] In step **432**, the DSDS device **240** transmits the session information to the CBRS core **210** via the CBSD **302**, which, in step **434**, the CBRS core **210** forwards to the DMTE **214**. In some implementations, steps **432** and **434** involve the same types of information as for step **324** of FIG. **3**.

[0045] In step **436**, the DMTE **214** analyzes the data received from the CBRS core **210** and selects duplicate sessions to be initiated (including, in this example, a duplicate session between the DSDS device **240** and the CBSD **302**. In step **438**, the DMTE **214** instructs the CBRS core **210** about those duplicate sessions to be initiated, and, in step **440**, the CBRS core **210** instructs the DSDS device **240** via the CBSD **302** to establish a communication session with the CBSD **302** that is a duplicate of the existing session with the gNodeB **304**.

[0046] In step **442**, the DSDS device **240** establishes that duplicate session with the CBSD **302**, and, in step **444**, merges that duplicate session with the existing session with the gNodeB **304**. In step **446**, the DSDS device **240** ceases to transmit data thereby terminating the existing session with the gNodeB **304**, while maintaining the newly restored session with the CBSD **302**. As before, in some possible implementations, the connection manager of the DSDS device **240** automatically terminates the session with the non-CBRS network after recognizing that existence of the two duplicate sessions. In other possible implementations, the DSDS device **240** transmits a message to the CBRS network confirming the establishment of the duplicate session with the non-CBRS network, and the CBRS network responds by instructing the DSDS device **240** to terminate the session with the non-CBRS network.

[0047] Here, too, the establishment of the duplicate session with the CBRS network and the termination of the session with the non-CBRS network all occur within the duration (e.g., 60 seconds) specified by the SAS **230** for modifying the communications of the CBRS network. In this way, the DMTE **214** pre-empts the conventional modification of CBRS network operations for DSDS devices by proactively handing off those communications back to the CBRS network, thereby restoring CBRS network operations to what they were before the incumbent's local activity occurred.

[0048] Although the present disclosure has been described in the context of the configuration **100** of FIG. **1** having a CBRS network and a non-CBRS network, those skilled in the art will understand that certain embodiments of the disclosure can be implemented in the context of two non-CBRS networks.

[0049] FIG. **5** is a simplified hardware block diagram of an example node **500** that can be used to implement any of the nodes of FIGS. **1-4**. As shown in FIG. **5**, the node **500** includes (i) communication hardware (e.g., wireless, wireline, and/or optical transceivers (TRX)) **502** that supports communications with other nodes, (ii) a processor (e.g., CPU microprocessor) **504** that controls the operations of the node **500**, and (iii) a memory (e.g., RAM, ROM) **506** that stores code executed by the processor **504** and/or data generated and/or received by the node **500**.

[0050] In certain embodiments, the present disclosure is a DMTE comprising a memory and at least one processor, coupled to the memory and operative to receive information that original operations of a first base station of a first wireless network are to be modified; determine that a DSDS device has a first active session with the first base station; and generate instructions for the DSDS device to establish a second active session with a second base station of a second wireless network in order to hand off the first active session with the first wireless network to the second

active session with the second wireless network, wherein the second active session is related to the first active session.

[0051] In at least some of the above embodiments, (i) the information is received from the first wireless network; (ii) the determining comprises the DMTE transmitting a request to the first wireless network for identification of any DSDS devices connected to the first base station; receiving identification of the DSDS device; transmitting a request to the first wireless network for identification of any active sessions between the DSDS device and the first base station; and receiving identification of the first active session; and (iii) the instructions are transmitted to the first wireless network.

[0052] In at least some of the above embodiments, the first wireless network is a Citizens Broadband Radio Service (CBRS) network and the second wireless network is a non-CBRS network.

[0053] In at least some of the above embodiments, the DMTE is further adapted to receive information that the original operations of the first base station are to be restored; determine that the DSDS device has the second active session with the second base station; and generate instructions for the DSDS device to establish a third active session with the first base station in order to hand off the second active session with the second wireless network to the third active session with the first wireless network, wherein the third active session is related to the second active session.

[0054] In certain embodiments, the present disclosure is a DSDS device comprising a memory and at least one processor, coupled to the memory and operative to establish a first active session with a first base station of a first wireless network; receive instructions to establish a second active session with a second base station of a second wireless network in order to hand off the first active session with the first wireless network to the second active session with the second wireless network, wherein the second active session is related to the first active session; establish the second active session; and terminate the first active session after establishing the second active session.

[0055] In at least some of the above embodiments, the DSDS device is adapted to automatically terminate the first active session.

[0056] In at least some of the above embodiments, before terminating the first active session, the DSDS device is adapted to inform the first wireless network of the establishment of the second active session and receive an instruction from the first wireless network to terminate the first active session.

[0057] In at least some of the above embodiments, the first wireless network is a CBRS network and the second wireless network is a non-CBRS network.

[0058] In at least some of the above embodiments, the DSDS device is further adapted to receive instructions to establish a third active session with the first base station in order to hand off the second active session with the second wireless network to the third active session with the first wireless network, wherein the third active session is related to the second active session; establish the third active session; and automatically terminate the second active session.

[0059] Unless explicitly stated otherwise, each numerical value and range should be interpreted as being approximate as if the word “about” or “approximately” preceded the value or range.

[0060] The use of figure numbers and/or figure reference labels in the claims is intended to identify one or more possible embodiments of the claimed subject matter in order to facilitate the interpretation of the claims. Such use is not to be construed as necessarily limiting the scope of those claims to the embodiments shown in the corresponding figures.

[0061] Although the elements in the following method claims, if any, are recited in a particular sequence with corresponding labeling, unless the claim recitations otherwise imply a particular sequence for implementing some or all of those elements, those elements are not necessarily intended to be limited to being implemented in that particular sequence. Likewise, additional steps may be included in such methods, and certain steps may be omitted or combined, in methods

consistent with various embodiments of the disclosure.

[0062] Reference herein to “one embodiment” or “an embodiment” means that a particular feature, structure, or characteristic described in connection with the embodiment can be included in at least one embodiment of the disclosure. The appearances of the phrase “in one embodiment” in various places in the specification are not necessarily all referring to the same embodiment, nor are separate or alternative embodiments necessarily mutually exclusive of other embodiments. The same applies to the term “implementation.”

[0063] Unless otherwise specified herein, the use of the ordinal adjectives “first,” “second,” “third,” etc., to refer to an object of a plurality of like objects merely indicates that different instances of such like objects are being referred to, and is not intended to imply that the like objects so referred-to have to be in a corresponding order or sequence, either temporally, spatially, in ranking, or in any other manner.

[0064] Also for purposes of this description, the terms “couple,” “coupling,” “coupled,” “connect,” “connecting,” or “connected” refer to any manner known in the art or later developed in which energy is allowed to be transferred between two or more elements, and the interposition of one or more additional elements is contemplated, although not required. Conversely, the terms “directly coupled,” “directly connected,” etc., imply the absence of such additional elements. The same type of distinction applies to the use of terms “attached” and “directly attached,” as applied to a description of a physical structure. For example, a relatively thin layer of adhesive or other suitable binder can be used to implement such “direct attachment” of the two corresponding components in such physical structure.

[0065] The described embodiments are to be considered in all respects as only illustrative and not restrictive. In particular, the scope of the disclosure is indicated by the appended claims rather than by the description and figures herein. All changes that come within the meaning and range of equivalency of the claims are to be embraced within their scope.

[0066] The functions of the various elements shown in the figures, including any functional blocks labeled as “processors” and/or “controllers,” may be provided through the use of dedicated hardware as well as hardware capable of executing software in association with appropriate software. Upon being provided by a processor, the functions may be provided by a single dedicated processor, by a single shared processor, or by a plurality of individual processors, some of which may be shared. Moreover, explicit use of the term “processor” or “controller” should not be construed to refer exclusively to hardware capable of executing software, and may implicitly include, without limitation, digital signal processor (DSP) hardware, network processor, application specific integrated circuit (ASIC), field programmable gate array (FPGA), read only memory (ROM) for storing software, random access memory (RAM), and non-volatile storage. Other hardware, conventional and/or custom, may also be included. Similarly, any switches shown in the figures are conceptual only. Their function may be carried out through the operation of program logic, through dedicated logic, through the interaction of program control and dedicated logic, or even manually, the particular technique being selectable by the implementer as more specifically understood from the context.

[0067] As will be appreciated by one of ordinary skill in the art, the present disclosure may be embodied as an apparatus (including, for example, a system, a network, a machine, a device, a computer program product, and/or the like), as a method (including, for example, a business process, a computer-implemented process, and/or the like), or as any combination of the foregoing. Accordingly, embodiments of the present disclosure may take the form of an entirely software-based embodiment (including firmware, resident software, micro-code, and the like), an entirely hardware embodiment, or an embodiment combining software and hardware aspects that may generally be referred to herein as a “system” or “network”.

[0068] Embodiments of the disclosure can be manifest in the form of methods and apparatuses for practicing those methods. Embodiments of the disclosure can also be manifest in the form of

program code embodied in tangible media, such as magnetic recording media, optical recording media, solid state memory, floppy diskettes, CD-ROMs, hard drives, or any other non-transitory machine-readable storage medium, wherein, upon the program code being loaded into and executed by a machine, such as a computer, the machine becomes an apparatus for practicing the disclosure. Embodiments of the disclosure can also be manifest in the form of program code, for example, stored in a non-transitory machine-readable storage medium including being loaded into and/or executed by a machine, wherein, upon the program code being loaded into and executed by a machine, such as a computer, the machine becomes an apparatus for practicing the disclosure. Upon being implemented on a general-purpose processor, the program code segments combine with the processor to provide a unique device that operates analogously to specific logic circuits. The term “non-transitory,” as used herein, is a limitation of the medium itself (i.e., tangible, not a signal) as opposed to a limitation on data storage persistency (e.g., RAM vs. ROM).

[0069] In this specification including any claims, the term “each” may be used to refer to one or more specified characteristics of a plurality of previously recited elements or steps. When used with the open-ended term “comprising,” the recitation of the term “each” does not exclude additional, unrecited elements or steps. Thus, it will be understood that an apparatus may have additional, unrecited elements and a method may have additional, unrecited steps, where the additional, unrecited elements or steps do not have the one or more specified characteristics.

[0070] As used herein, “at least one of the following: <a list of two or more elements>” and “at least one of <a list of two or more elements>” and similar wording, where the list of two or more elements are joined by “and” or “or”, mean at least any one of the elements, or at least any two or more of the elements, or at least all the elements. For example, the phrases “at least one of A and B” and “at least one of A or B” are both to be interpreted to have the same meaning, encompassing the following three possibilities: 1—only A; 2—only B; 3—both A and B.

[0071] All documents mentioned herein are hereby incorporated by reference in their entirety or alternatively to provide the disclosure for which they were specifically relied upon.

[0072] The embodiments covered by the claims in this application are limited to embodiments that (1) are enabled by this specification and (2) correspond to statutory subject matter. Non-enabled embodiments and embodiments that correspond to non-statutory subject matter are explicitly disclaimed even if they fall within the scope of the claims.

[0073] As used herein and in the claims, the term “provide” with respect to an apparatus or with respect to a system, device, or component encompasses designing or fabricating the apparatus, system, device, or component; causing the apparatus, system, device, or component to be designed or fabricated; and/or obtaining the apparatus, system, device, or component by purchase, lease, rental, or other contractual arrangement.

[0074] While preferred embodiments of the disclosure have been shown and described herein, it will be obvious to those skilled in the art that such embodiments are provided by way of example only. Numerous variations, changes, and substitutions will now occur to those skilled in the art without departing from the disclosure. It should be understood that various alternatives to the embodiments of the disclosure described herein may be employed in practicing the technology of the disclosure. It is intended that the following claims define the scope of the invention and that methods and structures within the scope of these claims and their equivalents be covered thereby.

Claims

1. A method comprising a dual-SIM, dual-subscription (DSDS) mobile transfer entity (DMTE): receiving information that original operations of a first base station of a first wireless network are to be modified; determining that a DSDS device has a first active session with the first base station; and generating instructions for the DSDS device to establish a second active session with a second base station of a second wireless network in order to hand off the first active session with the first

wireless network to the second active session with the second wireless network, wherein the second active session is related to the first active session.

2. The method of claim 1, wherein: the information is received from the first wireless network; the determining comprises: transmitting a request to the first wireless network for identification of any DSDS devices connected to the first base station; receiving identification of the DSDS device; transmitting a request to the first wireless network for identification of any active sessions between the DSDS device and the first base station; and receiving identification of the first active session; and the instructions are transmitted to the first wireless network.

3. The method of claim 1, wherein: the first wireless network is a Citizens Broadband Radio Service (CBRS) network; and the second wireless network is a non-CBRS network.

4. The method of claim 1, further comprising the DMTE: receiving information that the original operations of the first base station are to be restored; determining that the DSDS device has the second active session with the second base station; and generating instructions for the DSDS device to establish a third active session with the first base station in order to hand off the second active session with the second wireless network to the third active session with the first wireless network, wherein the third active session is related to the second active session.

5. A DMTE comprising: a memory; and at least one processor, coupled to the memory and operative to: receive information that original operations of a first base station of a first wireless network are to be modified; determine that a DSDS device has a first active session with the first base station; and generate instructions for the DSDS device to establish a second active session with a second base station of a second wireless network in order to hand off the first active session with the first wireless network to the second active session with the second wireless network, wherein the second active session is related to the first active session.

6. The DMTE of claim 5, wherein: the information is received from the first wireless network; the determining comprises the DMTE: transmitting a request to the first wireless network for identification of any DSDS devices connected to the first base station; receiving identification of the DSDS device; transmitting a request to the first wireless network for identification of any active sessions between the DSDS device and the first base station; and receiving identification of the first active session; and the instructions are transmitted to the first wireless network.

7. The DMTE of claim 5, wherein: the first wireless network is a Citizens Broadband Radio Service (CBRS) network; and the second wireless network is a non-CBRS network.

8. The DMTE of claim 5, wherein the DMTE is further adapted to: receive information that the original operations of the first base station are to be restored; determine that the DSDS device has the second active session with the second base station; and generate instructions for the DSDS device to establish a third active session with the first base station in order to hand off the second active session with the second wireless network to the third active session with the first wireless network, wherein the third active session is related to the second active session.

9. A method comprising a DSDS device: establishing a first active session with a first base station of a first wireless network; receiving instructions to establish a second active session with a second base station of a second wireless network in order to hand off the first active session with the first wireless network to the second active session with the second wireless network, wherein the second active session is related to the first active session; establishing the second active session; and terminating the first active session after establishing the second active session.

10. The method of claim 9, wherein the DSDS device automatically terminates the first active session.

11. The method of claim 9, wherein, before terminating the first active session, the DSDS device informs the first wireless network of the establishment of the second active session and receives an instruction from the first wireless network to terminate the first active session.

12. The method of claim 9, wherein: the first wireless network is a CBRS network; and the second wireless network is a non-CBRS network.

- 13.** The method of claim 9, further comprising the DSDS device: receiving instructions to establish a third active session with the first base station in order to hand off the second active session with the second wireless network to the third active session with the first wireless network, wherein the third active session is related to the second active session; establishing the third active session; and automatically terminating the second active session.
- 14.** A DSDS device comprising: a memory; and at least one processor, coupled to the memory and operative to: establish a first active session with a first base station of a first wireless network; receive instructions to establish a second active session with a second base station of a second wireless network in order to hand off the first active session with the first wireless network to the second active session with the second wireless network, wherein the second active session is related to the first active session; establish the second active session; and terminate the first active session after establishing the second active session.
- 15.** The DSDS device of claim 14, wherein the DSDS device is adapted to automatically terminate the first active session.
- 16.** The DSDS device of claim 14, wherein, before terminating the first active session, the DSDS device is adapted to inform the first wireless network of the establishment of the second active session and receive an instruction from the first wireless network to terminate the first active session.
- 17.** The DSDS device of claim 14, wherein: the first wireless network is a CBRS network; and the second wireless network is a non-CBRS network.
- 18.** The DSDS device of claim 14, wherein the DSDS device is further adapted to: receive instructions to establish a third active session with the first base station in order to hand off the second active session with the second wireless network to the third active session with the first wireless network, wherein the third active session is related to the second active session; establish the third active session; and automatically terminate the second active session.
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